

INTELLIGENT EDGE-AI MEDICINE DISPENSER

PROJECT BLISTER BOT — WEEK 1 TECHNICAL DOCUMENTATION & 3D MODEL REPORT

Project: Blister Bot (LLM Dispenser)

Platform: Arduino UNO Q (Linux + STM32)

Date: August 2026

1. Executive Summary & Project Vision

The **Intelligent Edge-AI Medicine Dispenser ('Blister Bot')** is an autonomous, fully offline healthcare assistant. Unlike conventional pill organizers that require manual sorting into individual bins or constant cloud connectivity, Blister Bot operates locally to interpret written prescriptions via optical character recognition (OCR), converse naturally with patients using a quantized Large Language Model (LLM), enforce strict overdose prevention rules, and **mechanically extract tablets directly from their original, uncut aluminum blister strips**.

2. Dual-Processor Hardware Architecture

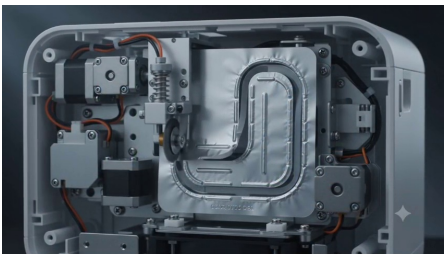
Subsystem Component	Hardware / Firmware	Primary Responsibilities & Workloads
Linux MPU	Qualcomm Dragonwing QRB2210	Runs localized LLM (llama.cpp + TinyLlama), speech recognition (Whisper.cpp), prescription OCR, face recognition, and database management.
Real-Time MCU	STM32U585 Microcontroller	Dedicated exclusively to non-blocking 2-axis CNC gantry motor control (AccelStepper), limit switches, safety interlocks, and plunger actuators.
Smart Display	7" Portrait TFT/LCD Panel	Renders patient dosage schedules, visual cross-check reference images, consumption graphs, and system telemetry.
Vision System	Integrated HD Camera Module	Handles patient facial recognition, prescription scanning, pill identification, coordinate mapping, and hand-to-mouth action verification.

3. Visual Reference Breakdown (From Video Analysis)

Key architectural and mechanical insights extracted from reference video frames:



Frame 002: Dual Chamber & Touch UI



Frame 005: CNC Gantry & U-Scoring

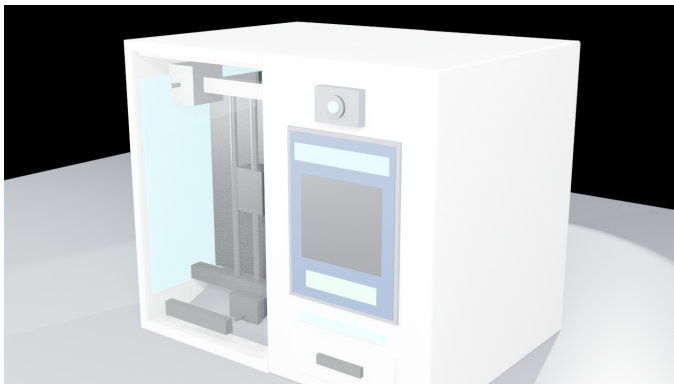


Frame 008: Appliance Overview

- **Dual-Chamber Layout:** The left chamber is illuminated and houses the multi-slot blister rack behind a transparent acrylic door; the right side houses the smart display, top AI camera, and pill output tray.
- **U-Shaped Foil Scoring:** To prevent pill damage caused by high-force punching, a motorized rotary cutter scores a 270-degree hinged flap in the aluminum backing, allowing a soft silicone plunger to push the pill out gently.

4. 3D Model Development & Blender Implementation

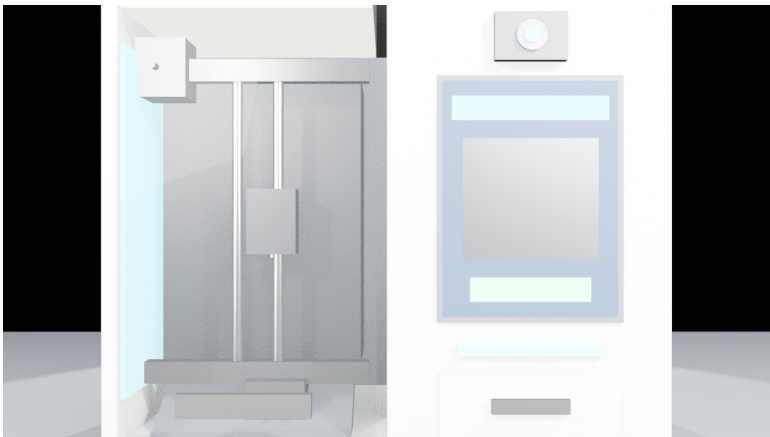
During Week 1, a complete, accurate 3D model of Blister Bot was procedurally constructed in **Blender 5.2 LTS** using Python automation script `create_blister_bot_model.py`. The model was saved as `blister_bot.blend` and rendered headlessly into high-resolution presentation preview images.



Render 1: Perspective Isometric View



Render 2: Deblistering Mechanism Close-Up



Render 3: Front Presentation View

5. Mechanical & Component Breakdown in 3D Model

- **Appliance Chassis & Enclosure:** Desktop form-factor with rounded matte white ABS plastic shell, dark metallic backing wall, top warm LED light strip (Mat_LED_Warm), cyan status light bar (Mat_LED_Cyan), and beveled transparent acrylic front window door.
- **Smart Touch Interface:** 7" portrait screen with customized UI widgets (pill schedule card, status bar, action buttons), top camera housing with glass lens, status LED strip, and lower dispensing cup drawer.
- **2-Axis CNC Deblistering Gantry:** Upper/lower X aluminum extrusions, vertical Y lead screw, linear guide rod, carriage block, and dual NEMA 17 stepper motors.
- **Scoring Toolhead & Ejection Pin:** Motorized rotary steel cutter disc engineered for 270-degree foil flap scoring and spring-loaded silicone plunger pin positioned above a funneled delivery chute.

6. Week 1 Deliverables & File Inventory

File / Resource Name	Type / Location	Description & Contents
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Project Blueprint.pdf	Week 1/	Full technical blueprint and software/hardware requirements specification.
blister_bot.blend	Week 1/	Native 3D project model file generated in Blender 5.2 LTS.
create_blister_bot_model.py	Week 1/	Python script for procedural geometry generation, shading, lighting, and rendering.
Week1_Documentation.md	Week 1/	Comprehensive markdown documentation report.
frames/	Week 1/frames/	Directory containing 10 reference image frames extracted from reference video.
renders/	Week 1/renders/	High-resolution 3D preview render images (perspective, mechanism, front).